

1 Introduction

Our notation:

- Random variable X_i with value x_i and possible values from $\mathcal{X}$,
- Random vector $\mathbf{X}$ with range $\mathcal{X}$ and some assignment $\mathbf{x}$,
- Set $\mathcal{A}$, and a measure $\mathbb{M}$.

2 Making Decisions under Uncertainty

3 Convergent Prior

Definition 3.1 (Probabilistic Model). We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of probability measures on $\mathcal{S}$.

Definition 3.2 (Divergence Measure). Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure* of $P \in \mathcal{P}$ for $\mathcal{X}$ as follows:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}_{X_i|E_i}(Q \parallel P) \right], \quad (1)$$

where $\mathbb{E}_Q[\cdot]$ denotes expectation relative to Q and each $\mathbb{D}_{X_i|E_i}(Q \parallel P)$ is the Kullback-Leibler divergence between $P(X_i|E_i)$ and $Q(X_i|E_i)$:

$$\mathbb{D}_{X_i|E_i}(Q \parallel P) = \sum_{x_i} Q(x_i | e_i) \ln \frac{Q(x_i | e_i)}{P(x_i | e_i)}.$$

Moreover, we define *the center of $\mathcal{M}$ for $\mathcal{X}$* to be a distribution $C_{\mathcal{X}} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_{\mathcal{X}} = \min_{P \in \mathcal{P}} \mathbb{D}_{\mathcal{X}}[P]. \quad (2)$$

Proposition. Using the linearity of expectation, we have:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(x_i, e_i) \ln \frac{Q(x_i | e_i)}{P(x_i | e_i)}. \quad (3)$$

Definition 3.3 (Parametric Model). Let θ be a random vector. A parametric model is a probabilistic model $\mathcal{M}_{\theta} \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability p , such that for every $s \in \mathcal{S}$ we have:

$$P(s) = \int \pi(s|\theta)p(\theta)d\theta, \quad (4)$$

while π is some known probability distribution.

Theorem 1. Let $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of random variable pairs in $\mathcal{M}_\theta$. For every $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{\mathcal{X}}[P] \leq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}_{X_i|e_i, \theta}(Q \parallel P) \mid \theta \right] \quad (5)$$

$$= \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}. \quad (6)$$

Proof. Let R be an arbitrary distribution in $\mathcal{M}_\theta$, since the Kullback-Leibler divergence is always positive, its expected value is also positive and for every parameter value θ we have:

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} \geq 0.$$

Using this, we can derive the following inequality for every distribution P and every parameter θ :

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}.$$

Since this inequality holds for any value of θ , the maximum of the right-hand side is also greater than or equal to the maximum of the left-hand side w.r.t θ :

$$\max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}. \quad (7)$$

$\mathcal{M}_\theta$ is a parametric model and $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$, while $r(\theta)$ is a probability distribution. Obviously for every θ we have:

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Since $r(\theta)$ is a distribution and is non-negative, we can multiply this inequality by $r(\theta)$ and integrate. Notice that the right hand side is not a function of θ and it will remain unchanged because $\int r(\theta) d\theta = 1$:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}_{X_i|e_i}(R \parallel P) \right] \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Now by Inequality 7, for *every* distribution P and R we conclude:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}_{X_i|e_i}(R \parallel P) \right] \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)},$$

which proves the theorem. $\square$

Corollary 1.1. *If a distribution $P \in \mathcal{P}$ satisfies:*

$$\sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{e} \in \mathcal{E}} \pi(\mathbf{x}, \mathbf{e} \mid \theta) \ln \frac{\pi(\mathbf{x} \mid \mathbf{e}, \theta)}{P(\mathbf{x} \mid \mathbf{e})} = \epsilon, \quad (8)$$

then $\mathbb{C}_{\mathcal{M}_\theta}[x_i \mid e_i] < \epsilon$, and P is an ϵ -convergent distribution for $x_i \mid e_i$.

Theorem 2 (Minimax [1]). *Let $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$ be compact convex sets. If $f : X \times Y \rightarrow \mathbb{R}$ is a continuous function that is concave-convex, i.e.*

- *$f(\cdot, y) : X \rightarrow \mathbb{R}$ is concave for fixed y , and*
- *$f(x, \cdot) : Y \rightarrow \mathbb{R}$ is convex for fixed x .*

Then we have that:

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y). \quad (9)$$

Theorem 3. *Let $\mathbf{X}$ and $\mathbf{E}$ be two random vectors in a parametric model $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$. The convergence measure of $x_i \mid e_i$ in $\mathcal{M}_\theta$ satisfies:*

$$\mathbb{C}_{\mathcal{M}_\theta}[x_i \mid e_i] < \max_{P \in \mathcal{P}} \sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{e} \in \mathcal{E}} \pi(\mathbf{x}, \mathbf{e} \mid \theta) \ln \frac{\pi(\mathbf{x} \mid \mathbf{e}, \theta)}{P(\mathbf{x} \mid \mathbf{e})} x e \theta, \quad (10)$$

where $P(\mathbf{x}, \mathbf{e}, \theta) = \pi(\mathbf{x}, \mathbf{e} \mid \theta)p(\theta)$.

Proof. □

Definition 3.4 (ϵ -convergent distribution). *Let S be a random variable, and $\mathcal{E}$ be a set of random variables in a probabilistic model $\mathcal{M}$. We say P is an ϵ -convergent distribution for $S \mid \mathcal{E}$ in $\mathcal{M}$, if for all $E \in \mathcal{E}$, $S \mid E$ ϵ -converges by P in $\mathcal{M}$.*

Proposition. *Let S be a random variable, and $\mathcal{E}$ be a set of random variables in a probabilistic model $\mathcal{M}$. If there exists $E \in \mathcal{E}$ such that $\mathbb{C}_{\mathcal{M}}[S \mid E] \geq \epsilon$, then there is no ϵ -convergent distribution for $S \mid \mathcal{E}$ in $\mathcal{M}$,*

Theorem 4. *Consider two probabilistic models $\mathcal{M}$ and $\mathcal{M}^*$. Let $\mathcal{E}$ be a countable set of random variables in $\mathcal{M}$, and E^* be a single random variable in $\mathcal{M}^*$. Suppose that there exists a random variable I in $\mathcal{M}^*$, such that for **every** probability measure P in $\mathcal{M}$, there exists a probability measure P^* in $\mathcal{M}^*$, such that for every $E_i \in \mathcal{E}$, every $s \in \mathcal{S}$ and every $e_i \in \mathcal{E}_i$, there exist some values s^*, e^* such that we have:*

$$P(S = s, E_i = e_i) = P^*(S^* = s^*, E^* = e^* \mid I = i). \quad (11)$$

In addition, I and E^ are conditionally independent:*

$$P^*(S^* \mid E^*) = P^*(S^* \mid E^*, I). \quad (12)$$

Then $S \mid \mathcal{E}$ ϵ -converges by P in $\mathcal{M}$, if $S^ \mid E^*$ ϵ -converges by P^* in $\mathcal{M}^*$.*

Proof. Assume, by contradiction, that $S|\mathcal{E}$ does not ϵ -converge by P in $\mathcal{M}$. Therefore, there must exist Q and $E_i \in \mathcal{E}$ in $\mathcal{M}$ such that:

$$\mathbb{E}_Q [\mathbb{D}(P_{S|E_i} || Q_{S|E_i})] \geq \epsilon .$$

By using equations 11 and 12 we can rewrite the left hand side:

$$\begin{aligned} \mathbb{E}_Q [\mathbb{D}(P_{S|E_i} || Q_{S|E_i})] &= \sum_{s \in \mathcal{S}} \sum_{e_i \in \mathcal{E}_i} \ln \frac{Q(s | e_i)}{P(s | e_i)} Q(s, e_i) \\ &= \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*, I = i)}{P^*(s^* | e^*, I = i)} Q^*(s^*, e^* | I = i) \\ &= \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^* | I = i) . \end{aligned}$$

Hence, we have:

$$\sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^* | I = i) \geq \epsilon .$$

Since $Q^*(i)$ is a distribution and is non-negative for every i , we can multiply both sides of this inequality to $Q^*(i)$ and sum over all values of i :

$$\begin{aligned} \sum_i \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^*, I = i) &\geq \sum_i \epsilon Q^*(i) \\ \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} \sum_i Q^*(s^*, e^*, I = i) &\geq \epsilon \sum_i Q^*(i) \\ \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^*) &\geq \epsilon , \end{aligned}$$

which means $S^*|E^*$ does not ϵ -converge by P^* in $\mathcal{M}^*$. $\square$

blahblahblah said Nobody [2]

Definition 3.5 (conditional model). Let $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$ be a probabilistic model. $\mathcal{M}$ conditioned on an event $\mathcal{A} \subset \mathcal{S}$ is a probabilistic model $\mathcal{M}_{|\mathcal{A}} \equiv (\mathcal{A}, \mathcal{P}_{|\mathcal{A}})$, where $\mathcal{P}_{|\mathcal{A}}$ is the set of probability measures obtained by conditioning every probability measure in $\mathcal{P}$ on $\mathcal{A}$.

Definition 3.6 (model radius). Let $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$ be a probabilistic model. We define the *radius* of $\mathcal{M}$ as follows:

$$\mathbb{R}_{\mathcal{M}} = \min_{P \in \mathcal{P}} \max_{Q \in \mathcal{P}} \{\mathbb{D}_{\text{KL}}(Q || P)\} = \min_{P \in \mathcal{P}} \max_{Q \in \mathcal{P}} \sum_{s \in \mathcal{S}} Q(s) \ln \frac{Q(s)}{P(s)} . \quad (13)$$

References

- [1] Ding-Zhu Du. *Minimax and Its Applications*, pages 339–367. Springer US, Boston, MA, 1995.
- [2] Nobody Jr. My article, 2006.